INCH-POUND
MIL-PRF-49166B(CR)
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SUPERSEDING
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28 May 1987

PERFORMANCE SPECIFICATION
MECHANICAL SCANNER, MX-9872(V)/UA;
GENERAL SPECIFICATION FOR

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the general attributes of the common module, Mechanical Scanner, MX-9872(V)/UA. This specific coverage of each variation of the MX-9872(V)/UA are contained in specification sheets attached as supplements to this specification.

2. APPLICABLE DOCUMENTS (See supplement 1 for list of specification sheets.)

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification whether they are listed or not.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - Standard general requirements for electronic equipment.

(Unless otherwise indicated, copies of the above handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC-S-815A - General requirements for soldering electronic inter-connections.

(Application for copies should be addressed to The American National Standards Institute, Inc., 1430 Broadway, N.Y., N.Y. 10018)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheet. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

3.3 Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.4 Description. The Mechanical Scanner, MX-9872(V)/UA, herein known as the Scanner, normally consists of a scan mechanism assembly and housing. The scan mechanism includes a flat scan mirror, interlace gimbal, and associated drive components. The scan mirror pivots about the scan axis and oscillates about the interlace axis. The Scanner has provisions for mounting a phase lens and counterweight for system unique requirements.

3.4.1 Detailed requirements. The detailed requirements for the scanner shall be as specified in the applicable specification sheet (see 3.1).

3.4.2 Weight. The weight shall be as specified in the applicable specification sheet (See 3.1).

3.5 Components. The Scanner consists of the following major components:

a. Scan mirror assembly
b. Solenoid (2)
c. Interlace flex pivot assembly (2)
d. Scan transducer assembly
e. Interlace transducer assembly
f. Upper scan flex pivot assembly
g. Lower scan flex pivot assembly
h. Return spring assembly (4)
I. Scanner housing
j. Scanner housing cover
k. Gimbal
l. Wiring harness
m. Gimbal counterweight

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3.6 Performance characteristics. (See 3.1.)

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Scan frequency of the scanner shall be variable from 20 IR frames/40 fields to 62 IR frames/124 fields per second. For variable scan frequency measurement and verification, see 4.5.1.1.

3.6.1.2 Specific scan frequency (60 Hz). (See 3.1.)

3.6.1.3 Synchronous scan frequency. (See 3.1.)

3.6.1.4 Scan angle. (See 3.1.)The scan angle shall be determined as follows:

a. The test apparatus shall consist of a holding fixture, a projection system capable of projecting a spot or marker into the scanner along the x-axis (Figure 4) and an appropriate scale graduated in degrees measured about the scan mirror axis with its zero position located on the y-axis (Figure 4).

b. The Scanner shall be operated and the angular excursions of the projected spot in each direction shall be recorded. One-half of the sum of the excursions shall be the measured scan angle.

3.6.2 Interlace. The Scanner shall operate in a 2:1 or 1:1 interlace mode and shall interlace either once per azimuth scan or once every other azimuth scan, respectively. The interlace switching time defined in Figure 1 shall be as specified in the applicable specification sheet (see 3.1). Verify specified performance through the use of 4.5.2.

3.6.3 Scan direction. The Scanner shall operate in a bi-directional mode at 2 scans per cycle and shall display the waveform shapes and phase relationships as specified in the applicable specification sheet (see 3.1). Verify specified performance through the use of 4.5.3.

3.6.4 Power consumption. Power required to operate the Scanner and the Scan-Interlace module is dependent upon the specific Scanner configuration but shall not exceed 7.5 watts (See applicable specification sheet (see 3.1) for specific power consumption.) Verify specified performance through the use of 4.5.4.

3.6.5 Scan efficiency. The scanner scan efficiency shall be as noted in the applicable specification sheet (see 3.1). Verify specified performance through the use of 4.5.5.

3.6.6 Scan jitter in azimuth. The amount of jitter measured in the scan direction shall not be greater than 0.75 milliradian as determined by using an oscilloscope. Verification of specified performance shall be demonstrated though use of 4.5.6.

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3.6.7 Scan jitter in interlace. The interlace amplitude shall be repeatable to within a ±10% deviation of the height of a display element as measured over the center 80% of the azimuth field of view. Verify specified performance through the use of 4.5.7.

3.6.8 Modulation transfer function (MTF). The MTF of each side of the scan shall be no less than 95 % at 0.667 cycles per milliradian (cy/mrad). Verify specified performance through the use of 4.5.8.

3.7 Environmental conditions.

3.7.1 Temperature shock. The scanner shall not be damaged (see 6.4.1) by sudden changes in temperature between -54° and +95°C (see 6.2.f).

3.7.2 High temperature. The Scanner shall not be damaged by storage to +95°C or continuous operation to +71°C, and shall operate for 30 minutes at +95°C. The average interlace waveform amplitude (Figure 1) shall not change more than ±20 percent from the initial setup at ambient temperature (see 6.4.3) over the specified high temperature operating range (see 6.2.f).

3.7.3 Low temperature. The Scanner shall not be damaged by operation to -54°C or storage to -62°C. After soaking at -62°C, the average interlace waveform amplitude (Figure 1) shall not change more than ±20 percent from initial setup at ambient temperature (see 6.4.3) when stabilized at -30°C (see 6.2.f).

3.7.4 Shock. The Scanner shall not be damaged by high intensity shocks of 80 g's peak amplitude with 11 milliseconds duration and by bench handling tests. The average interlace waveform amplitude (Figure 1) shall return to the initial amplitude within 0.2 second after the Scanner is subjected to the operating shock environment, not to include bench handling test (see 6.2.f).

3.7.5 Vibration. The Scanner shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes as shown in Figure 3 for the x, y, and z axes defined in Figure 4. During specified vibration, the average interlace waveform amplitude (Figure 1) shall not deviate from the amplitude without vibration by more than ±50 percent. The sinusoidal cycle time shall be 120 minutes per axis The dwell time shall be 1/6 of the cycling time at each resonance (see 6.2.f).

3.8 Soldering. Electrical soldering shall conform to ANSI/IPC-S-815A.

3.9 Reliability. The Scanner shall have a mean time between failure (MTBF) of at least 6,058 hours θ.

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3.10 Nameplates and product marking. Unless otherwise specified, Scanner parts and components, subassemblies, and assemblies shall be marked for identification and reference designation markings in accordance with the requirements of the contract or purchase order. Use requirement 67 of MIL-HDBK-454 for guidance.

3.11 Treatment and painting. Unless otherwise specified, the Scanner and its components and parts shall have a surface treatment which shall prevent rust, corrosion, and fungus growth in the full range of environmental requirements. External color and surface shine shall reduce light reflection and reduce recognition. Adherence must be assured through the full range of environmental and use factors.

3.12 Preconditioning. Each Scanner shall be operated for at least 8 hours prior to Group A inspection.

3.13 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires or integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one or more complete Scanners when a first article sample is required (see 3.1). This inspection shall include the inspection of 4.2.1 and the tests of 4.2.2.

4.2.1 Inspections. All mechanical scanners shall be examined for conformance with the form, fit, function and interface requirements of the applicable specification sheet (see 3.1). Inspections shall be made using Table I and sampling procedures as directed by the contract or purchase order.

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TABLE I. Visual Inspections.

Inspection Requirement	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.5	3.4
Weight not as specified	3.4.2	See (3.1)
Soldering not as specified	3.8	ANSI/IPC S-815A
Marking not as specified	3.10	See contract or purchase order
Workmanship not as specified	3.13	3.12
Treatment and painting not as specified	3.11	Manufacturer's Internal Procedure

4.2.2 Tests. Upon successful completion of inspections specified in 4.2.1, perform the tests of Table II in any order, except that the modulation transfer function (MTF) test shall follow the high temperature test. Failure of any tests shall constitute failure of first article inspection.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Variable scan frequency	3.6.1.1	4.5.1.1
Specific scan frequency	3.6.1.2	4.5.1.2
Synchronous scan frequency	3.6.1.3	4.5.1.3
Scan angle	3.6.1.4	4.5.1.4
Interlace	3.6.2	4.5.2
Scan direction	3.6.3	4.5.3
Power consumption	3.6.4	4.5.4
Scan efficiency	3.6.5	4.5.5
Scan jitter in azimuth	3.6.6	4.5.6
Scan jitter in interlace	3.6.7	4.5.7
Modulation transfer function	3.6.8	4.5.8
Temperature shock	3.7.1	4.6.1
High temperature	3.7.2	4.6.2
Low temperature	3.7.3	4.6.3
Shock	3.7.4	4.6.4
Vibration	3.7.5	4.6.5
Reliability	3.9	4.8

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4.2.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities.

4.3 Conformance inspection.

4.3.1 Inspection. Visual inspection shall be as specified in 4.2.1.

4.3.2 Tests.

4.3.2.1 Group A inspection. Group A inspection shall be conducted at ambient temperature on all scanners following the preconditioning of 3.11 and the visual inspection of 4.3.1. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Specific scan frequency	3.6.1.2	4.5.1.2
Synchronous scan frequency	3.6.1.3	4.5.1.3
Scan angle	3.6.1.4	4.5.1.4
Interlace	3.6.2	4.5.2
Power consumption	3.6.4	4.5.4
Scan efficiency	3.6.5	4.5.5
Scan jitter in azimuth	3.6.6	4.5.6
Scan jitter in interlace	3.6.7	4.5.7

4.3.2.2 Group B inspection. Not required.

4.3.2.3 Group C inspection. Group C inspections shall be conducted on scanners selected from units which have passed the tests in 4.3.2.1. The sample(s) shall be tested in accordance with the inspections listed in Table IV. Group C tests may be performed in any order, except that the modulation transfer function test **(**4.4.2**)** shall be performed last.

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TABLE IV. Group C Inspection.

Inspection	Requirement paragraph	Test paragraph
Scan direction	3.6.3	4.5.3
High temperature	3.7.2	4.6.2
Low temperature	3.7.3	4.6.3
Shock	3.7.4	4.6.4
Vibration	3.7.5	4.6.5
Modulation transfer function	3.6.8	4.5.8

4.3.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract (see 6.2).

4.3.2.3.3 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the tests of Table III.

4.3.2.4 Group D inspection. Not required.

4.4 Sampling. Sampling for Group A, and C shall be in accordance with the contract or purchase order.

4.5 Methods of inspection. Unless otherwise specified, the tests shall be performed utilizing the Scan-interlace module as specified in the applicable specification sheet.

4.5.1 Scan angle and frequency.

4.5.1.1 Variable scan frequency. The scanner shall be connected to Scan-Interlace, PL-1408/UA and mounted on a suitable fixture. Power shall be applied to the scanner and the output of the azimuth transducer (Scan-Interlace) shall be monitored with a calibrated frequency counter. The output of the scan drive electronics shall be varied to demonstrate that the requirements of 3.5.1.1 are met. Failure to meet the requirements of 3.5.1.1 shall constitute failure of this test.

4.5.1.2 Specific scan frequency. See 3.1.

4.5.1.3 Synchronous scan frequency. See 3.1.

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4.5.1.4 Scan angle. The angle shall be determined as follows:

4.5.1.4.1 Test apparatus. The test apparatus shall consist of a holding fixture, a projection system capable of projecting a spot or marker into the scanner along the x-axis (Figure 4) and an appropriate scale graduated in degrees measured about the scan mirror axis with its zero position located on the 7-axis (Figure 4).

4.5.1.4.2 Test conduction. The scanner shall be operated and the angular excursions of the projected spot in each direction shall be recorded. One-half of the sum of the excursions shall be the measured scan angle.

4.5.1.4.3 Failure criteria. Failure to meet the requirements of 3.5.1.4 shall constitute failure of this test.

4.5.2 Interlace. With the scanner operating in a suitable fixture, the output of the interlace position transducer shall be monitored with an oscilloscope. The waveform switching times (Figure 1) shall be measured to verify the requirement of 3.5.2 is met. Failure to meet requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Scan direction. With the scanner operating, the outputs of the azimuth and interlace position transducers or the interlace transducer and scan tachometer (using the appropriate scan-interlace module) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships as specified in the applicable specification sheets. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Power consumption. The scan drive electronics shall be connected to constant voltage power supplies. The current being drawn by the electronics from each voltage source shall be measured to obtain the average current being drawn. The power delivered to the module shall be calculated as the sum of the products of the power supply voltages and their average currents. Failure to meet requirements of 3.5.4 shall constitute failure of this test.

4.5.5 Scan efficiency. With the scanner operating, the period of the azimuth position transducer waveform shall be measured with an oscilloscope. The scanner shall then be stopped, and the scan mirror moved from contact with one return spring to contact with the opposite return spring. Simultaneously, the signal level of the transducer at the spring contact points shall be noted on the oscilloscope. With the scanner operating, the period of time that the transducer output is between the two signal levels during one cycle shall be determined from the oscilloscope display. The ratio of this period of time to the azimuth position transducer waveform period which was measured previously is the duty cycle or scan efficiency of the scanner operated with the appropriate scan-interlace module. See applicable specification sheet (See 3.1) when a scan-interlace module other than PL-1408/UA is used. Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

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4.5.6 Scan jitter in azimuth. The scanner shall be set up and operating as in 4.5.1.2. Using an oscilloscope, the scan position zero crossing detection circuit shall be measured for jitter on any positive transition edge (see Figure 2). Failure to meet requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Scan jitter in interlace. The output of the interlace position transducer shall be monitored with an oscilloscope. The average interlace movement signal, defined as the average of the difference between the waveform amplitude on the forward scan and the waveform amplitude on the reverse scan at identical scan mirror azimuth positions, shall be measured. Scan jitter in interlace is defined as the variation of the above differences from the average interlace movement signal. Failure to meet requirements of 3.5.7 shall constitute failure of this test.

4.5.8 Modulation transfer function (MTF). The scanner shall be located in a 38.0, ±0.5 mm collimated bundle of visible radiation (660 nanometers peak, with 20 nanometer half-power bandwidth) parallel and centered on the optical axis. The scanner shall be tested on the MTF apparatus that yields sine wave response. The image area to be analyzed shall be at least ±1.5 milliradians around the optical axis. The degradation due to the scanner shall be determined by comparison of the MTF measurements with the test apparatus MTF.

4.5.8.1 Test apparatus MTF. The MTF for the test apparatus shall be determined by rotating the optical imaging system to be aligned to the collimated beam. The maximum MTF of the test apparatus shall be determined by an optimization search of the image.

4.5.8.2 MTF. The MTF of the scanner shall be measured in both azimuth and elevation directions for each mirror surface and there shall be no refocusing allowed between the azimuth and elevation requirements. In each direction and for each side, the ratio of the scanner MTF at 0.667 cycles/milliradian to the corresponding MTF measurement of the test apparatus (see 4.5.8.1) shall be obtained. Failure to meet the requirement of 3.5.8 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing shall be performed in accordance with the requirements of this specification, except when a contractor's own test plan showing equivalent methods has been approved by the Government. The operating tests required by this specification conducted before , during and after each environmental test shall be those listed in Table III herein.

4.6.1 Temperature shock. The scanner shall be tested in a manner similar or equal to 6.3, except that the temperature of Step 2 shall be -54°C and the temperature of Step 1 and Step 4 shall be +95°C. The scanner shall not be operated during temperature shock tests (no measurements shall be taken during tests). Failure to meet requirements of 3.6.1 shall constitute failure of this test.

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4.6.2 High temperature. The scanner shall be tested in a manner equal or similar to 6.3 except that the temperature of step 2 shall be +95°C and the scanner shall be operated during the final 30 minutes of Step 3 (measurements shall not be taken during Step 3). The temperature of Steps 4 and 5 shall be +71°C. The high temperature test criteria specified in 4.5.1.2 and the average interlace waveform amplitude shall be measure din place of the table III tests during high temperature tests. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Low temperature. The scanner shall be tested in a manner equal or similar to 6.3 except that the storage temperature of Step 2 shall be -62°C, the low operating temperature of Step 4 shall be -54°C and the low operating temperature for the interlace waveform criterion shall be interlace waveform amplitude shall be measured in place of the Table III tests during low temperature tests. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Shock. The scanner shall be tested in a manner equal or similar to 6.3 (100 g's peak value at 11 milliseconds time duration) and Procedure V. During procedure IV, the scanner shall be operated and the average interlace waveform amplitude shall be measured in place of Table III. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Vibration. The scanner shall be tested in a manner equal or similar to 6.3 except that the curve shown in Figure 3 of this specification shall be used in lieu of the curve. The sinusoidal cycle time shall be 120 minutes per axis. The dwell time shall be 1/6 of the cycling time at each resonance. The scanner shall be operated during vibration and the average interlace waveform amplitude shall be measured in place of the Table III test. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

4.7 Reliability.

4.7.1 First article reliability. The lower test MTBF (θ) shall be demonstrated in accordance with the following subparagraphs:

4.7.1.1 Test level.

a. Temperature +50°C, (+5 -0)°C

b. Temperature cycling - None

c. Vibration - 2.2 g, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be in the direction of the x-axis (see Figure 3).

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d. Power on-off cycling - Turn power on and let temperature stabilize, hold for 3 hours, then turn power off and let temperature stabilize. This cycle shall continue throughout the test.

4.7.1.2 Parameters. The discrimination ratio θ_o/θ_1 shall be 3.

4.7.1.3 Failure criteria and test length. The tests of Table III shall be performed at least once prior to and once after the tests of 4.7.1.1. Additional testing of Table III may be selected by the contractor. The results of all Table III tests and any physical damage shall be used to determine the success or failure of the reliability test. In the event of a Table III test failure, the time interval between the preceding successful test and the failure shall not contribute to the total cumulative hours. Testing shall continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with Figure 5. Only equipment "on" time shall contribute to the total unit hours. No single equipment's "on" time shall contribute to the total unit hours. No single equipment's "on" time shall be less than one-half the average operating time of all equipment on test.

4.8 Preconditioning. The scanner shall be mounted on a suitable fixture and connected to a scan-interlace module or similar circuit. Power shall be applied and the scanner shall be operated for at least 8 hours. Monitoring is not required during preconditioning.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory in the performance of the contract.)

6.1 Intended use. The Scanner is intended for infrared systems. When the scanner is mounted in an infrared (IR) system, the front side of the scan mirror directs incoming IR energy through a set of IR imaging optics onto an array of IR detectors. The back side of the mirror directs the visual output from a light emitting diode (LED) array through a set of visible collimating optics and into a visible display optics assembly.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see5.1).

d. If First article tests are required.

1) Time frame for submission of first article test reports when first article tests are required.

2) Time frame for approval of first article test reports when first article tests are required.

e. Production delivery schedule.

1) Defined in terms of monthly lots.

2) Actions required in case of Group C failures.

f. For environmental testing, use 6.3 as a possible source (see 6.3).

g. If Scanners are procured as part of end item assemblies, temperature shock, shock and vibration should not be required.

h. When ordering a specific type of scanner, only those parameters in each test group that are applicable to that scanner should be required.

i. Scan-Interlace PL-1403/UA and PL-1403A/UA per DLSM-A-657598,and DLSM-A-771731, respectively, and/or Scan-Interlace PL-1408/UA per DLSM-B-773232 may be ordered from any existing supplier of these modules.

j. Group C inspection monthly sample size to be established based on cumulative totals of concurrent contracts.

k. Marking requirements should be specified.

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6.3 Environmental inspection procedures. (see Section 3.6). The test methods and procedures of MIL-STD-810 listed below are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the material to be tested.

Inspection	Spec. Paragraph	MIL-STD-810 Method	MIL-STD-810 Procedure	Remarks
Temperature shock	3.7.1	503	I	Temp. of Step 2 shall be -54°C; temp. of Step 1 and Step 4 shall be +95°C.
High temperature	3.7.2	501	I	Storage temp. of Step 2 shall be -62°C; the low operation temp of Step 4 shall be -54°C; the low operation temp. for the interlace waveform criterion shall be -30°C.
Shock	3.7.4	516.1	IV, V	Use Figure 5, 16.1.1 (80 g peak value @ 11 milliseconds time duration and procedure V. During procedure IV, the scanner shall be operated and the average interface waveform amplitude shall be measured in lieu of Table III.
Vibration	3.7.5	514.1	I, VIII	Use the curve shown in Figure 3. The scanner shall be operated during vibration and the average interface waveform shall be measured in lieu of Table III.

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6.4 Definitions.

6.4.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of scanner performance to values less than specified herein.

6.4.2 Failure. A noted incapability to properly attain a specified performance requirement during the conduct of a test or examination process described within this specification.

6.4.3 Ambient temperature. Ambient temperature is +23°C, ±5°C.

6.5 Subject term (key word) listing.

Scan mechanism
Scan-Interlace

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5855-0120)

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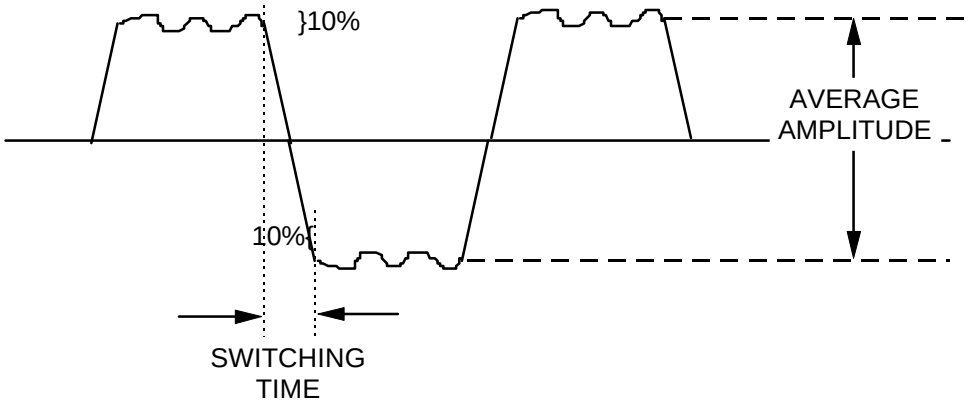


FIGURE 1. INTERLACE TRANSDUCER OUTPUT.

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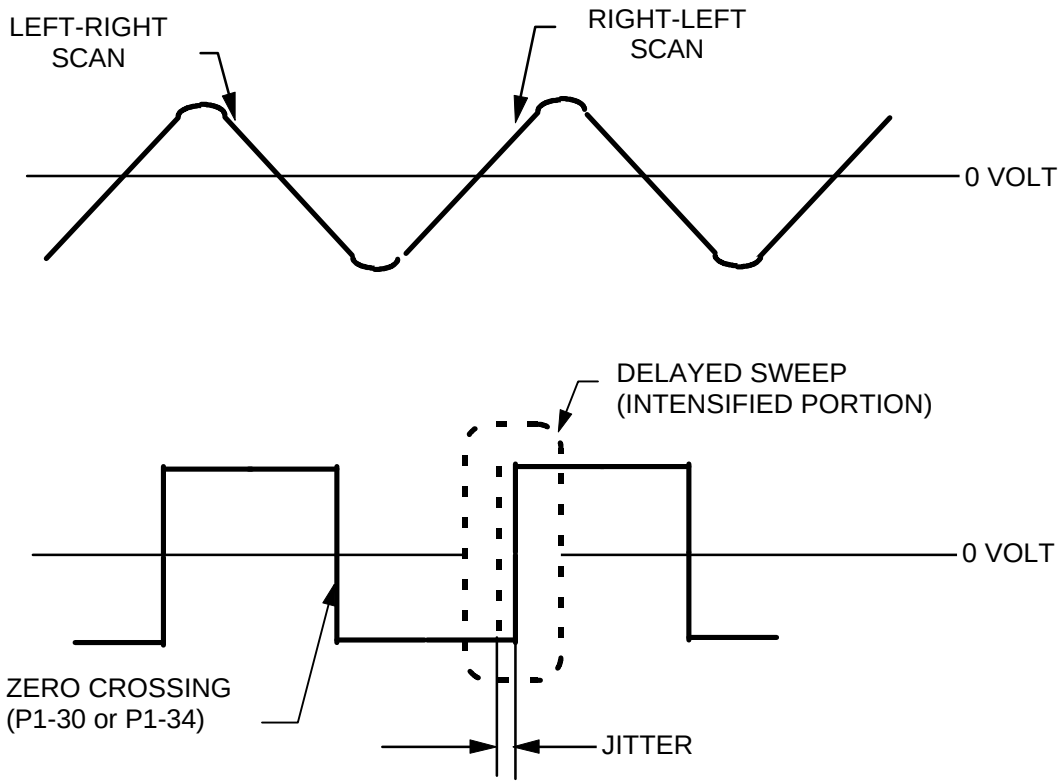


FIGURE 2. ZERO CROSSING WAVEFORM JITTER.

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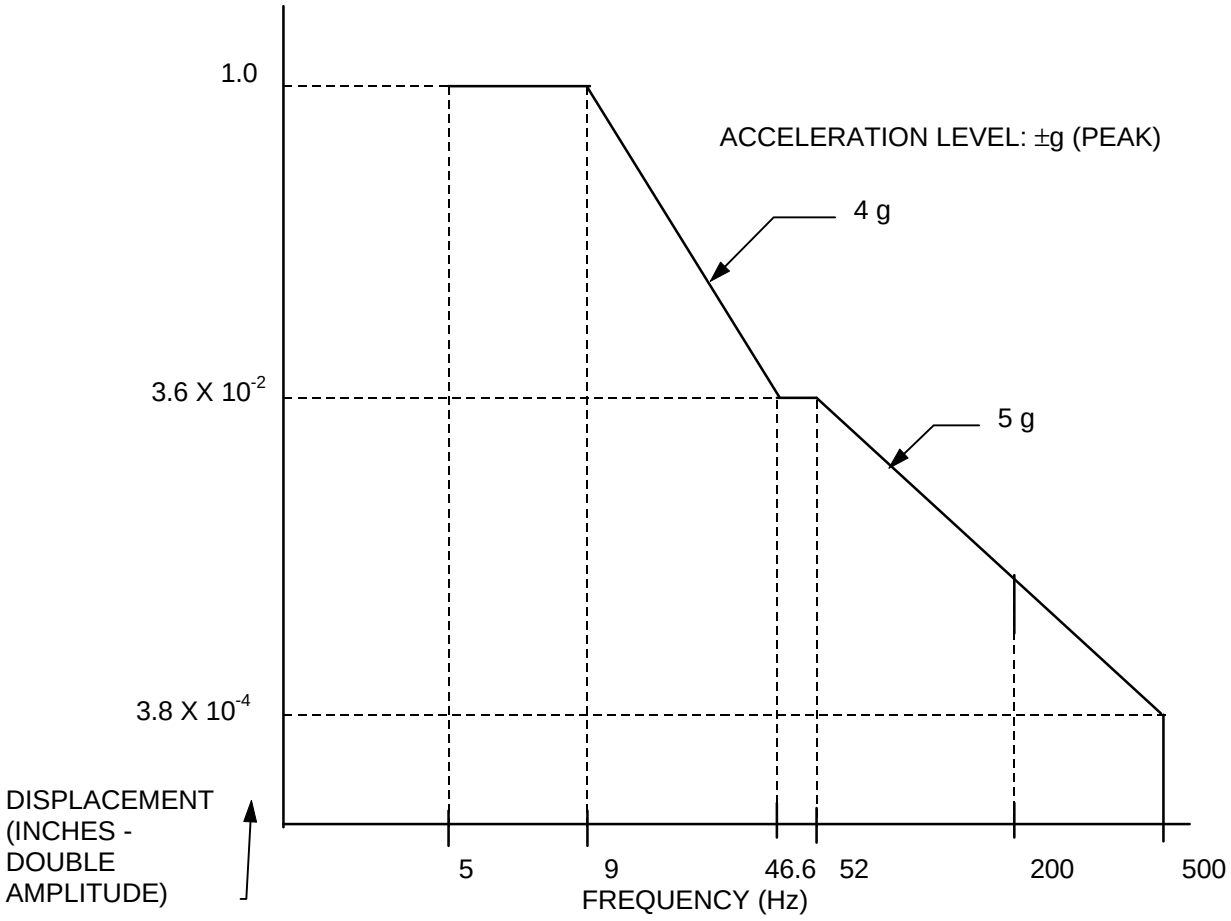


FIGURE 3. VIBRATION TEST PROFILE .

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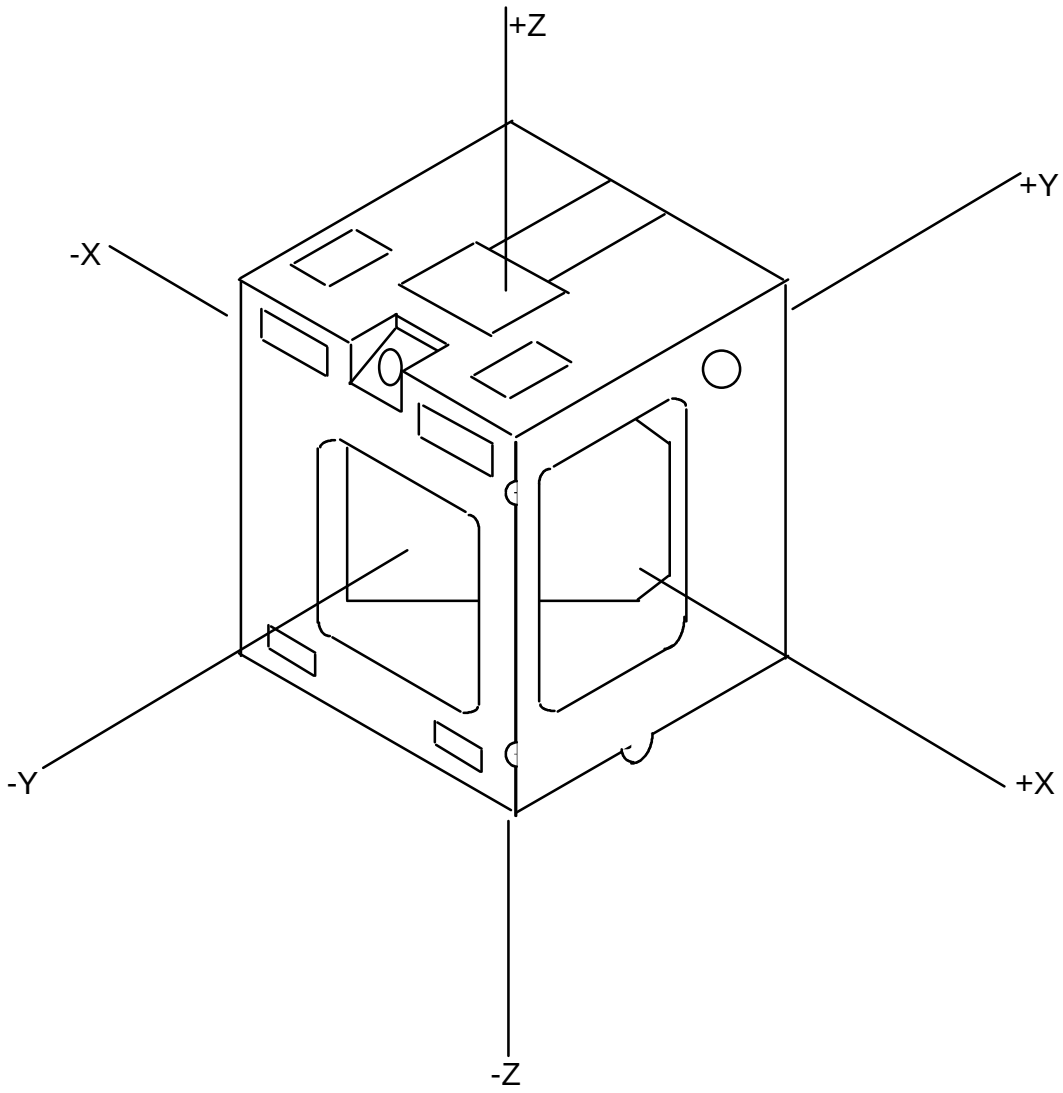
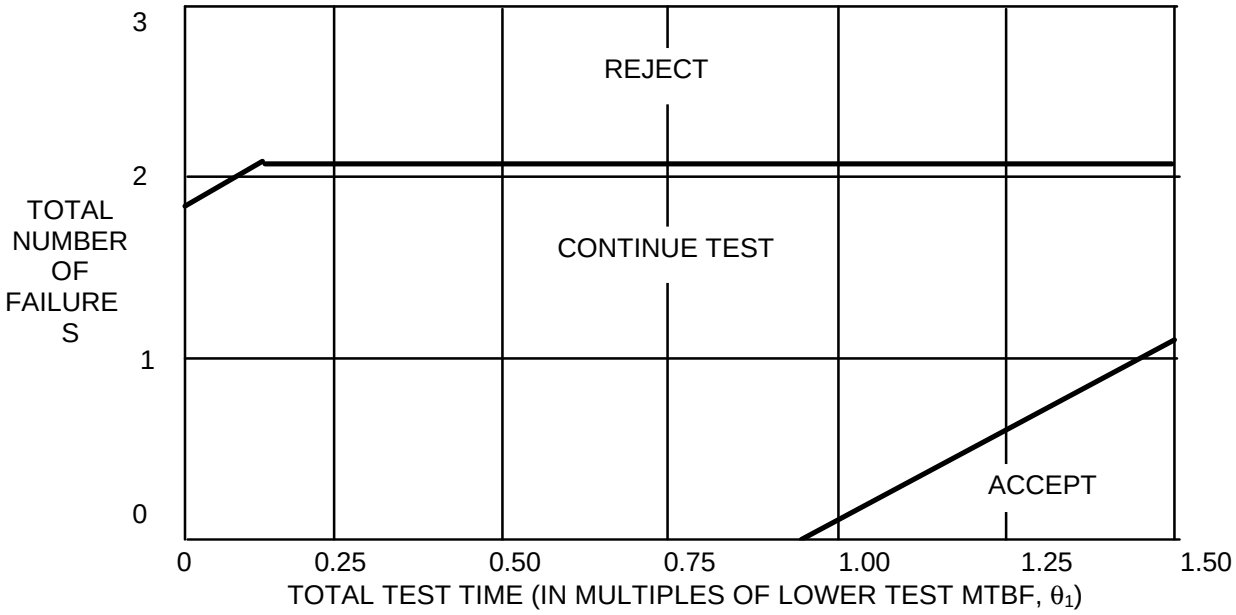


FIGURE 4. SCANNER GEOMETRIC REFERENCES.

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TOTAL TEST TIME*

NUMBER OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF EQUIPMENT “ON TIME” AND IS EXPRESSED IN MULTIPLES OF LOWER TEST MTBF θ_1

FIGURE 5. FIRST ARTICLE RELIABILITY, ACCEPT-REJECT CRITERIA.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49166B(CR)	**2. DOCUMENT DATE (YYMMDD)** 27 August 1997

3. DOCUMENT TITLE: Mechanical Scanner, MX-9872(V)/UA; General Specification for

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9104 (2) AUTOVON 992-9104
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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